

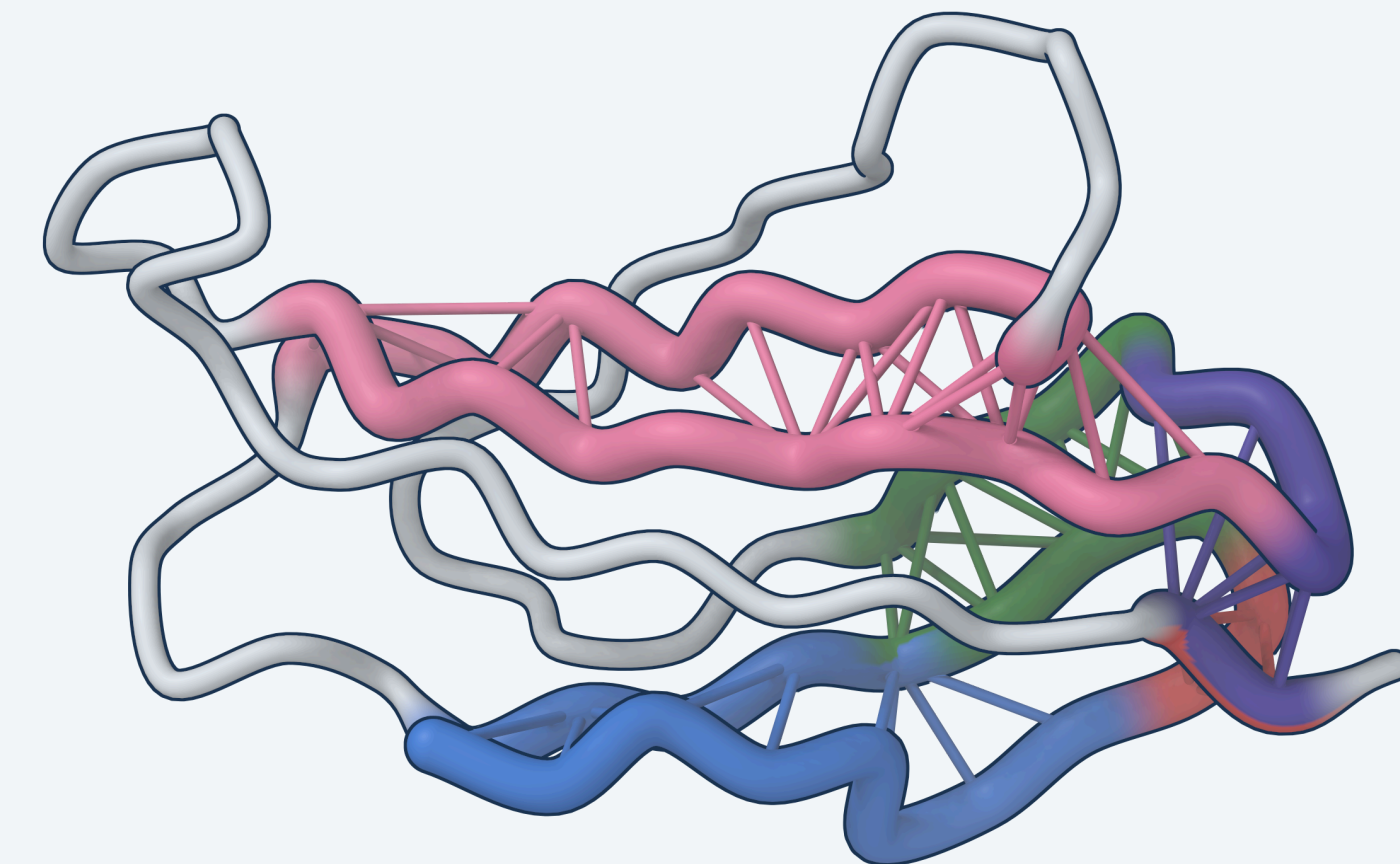
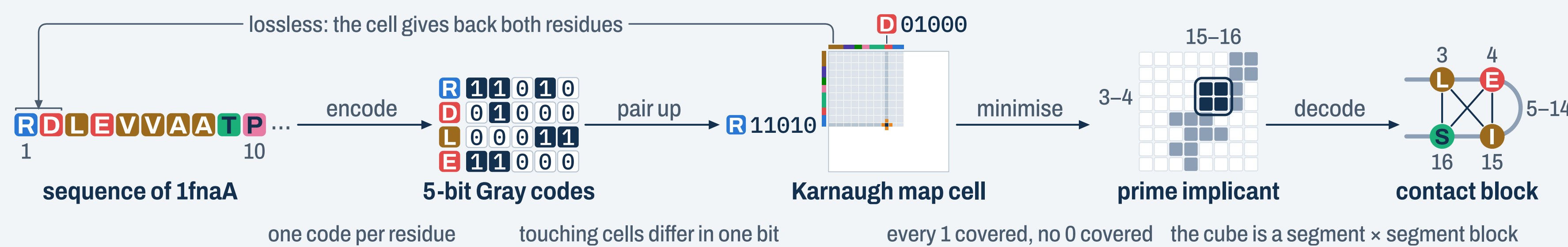
Is KMAP Possible? Formal Verification of Karnaugh Map Encoding for Biological Sequences in Lean 4

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000 The finding

We formalised in Lean 4 the mathematics that turns DNA, RNA and protein sequences into Karnaugh maps: 131 machine-checked theorems, none left unproved. On this foundation every structure becomes an exact Boolean circuit whose rules hold without exception, for proteins, a viral species and folding simulations.

Gray-coded grids of sequences, from Petoukhov's genetic Karnaugh matrices¹ to chaos-game maps (FCGR)^{2,3}, are usually taken on trust. Here the encodings are theorems and every circuit and rule is checked on every case.



Fibronectin type-III domain (PSICOV 1fnaA, 91 residues). Its exact contact circuit, read as rules: block rules that continue one another are strand pairs, one colour each.

Strand pairs, found as blocks

- 25–34 with 61–70: 6 blocks
- 1–8 with 14–18: 4 blocks
- 11–14 with 53–56: 3 blocks
- 57–60 with 89–90: 2 blocks
- 9–10 with 89–90: 1 block

Turn any protein into a circuit

00 01 11 10



Any PDB entry, step by step, to its rules and inferences.

001 A code that changes one bit at a time

Binary counting can flip many bits at once: 0111 $\rightarrow$ 1000 flips four. The reflected Gray code $g(n) = n \oplus (n \gg 1)$ visits every number once and changes exactly one bit per step. DNA needs two bits, and A C G T is itself a Gray cycle.

n	binary	flips	Gray $g(n)$	flips
0	000		000	
1	001	1	001	1
2	010	2	011	1
3	011	1	010	1
4	100	3	110	1
5	101	1	111	1
6	110	2	101	1
7	111	1	100	1

DNA needs two bits

00 A 01 C

10 T 11 G

the bit that flips

011 A Karnaugh map⁴ is a folded truth table

Order a grid's rows and columns by the Gray code and touching cells differ in one variable, even across the edges, which wrap like a torus. Put a k -mer's first letters on the rows and the rest on the columns: each 2-mer owns one cell of a 4×4 map, each 4-mer one of a 16×16 map.

first base	second base	
	A 00 C 01 G 11 T 10	
A 00	AA AC AG AT	00 10
C 01	CA CC CG CT	01 11
G 11	GA GC GG GT	11 10
T 10	TA TC TG TT	

One cell per k -mer
4 $\times$ the cells per base

k = 1 k = 2 k = 3 k = 4

2 $\times$ 2 4 $\times$ 4 8 $\times$ 8 16 $\times$ 16

010 Amino acids on a five-dimensional cube

Proteins need five bits: 32 codewords for 20 residues. Numbering residues group by group before Gray-coding keeps chemistry close. All 190 pairwise distances are proved: 40 pairs sit one bit apart, only four sit five apart, and the code uses exactly half of the cube's 80 edges.

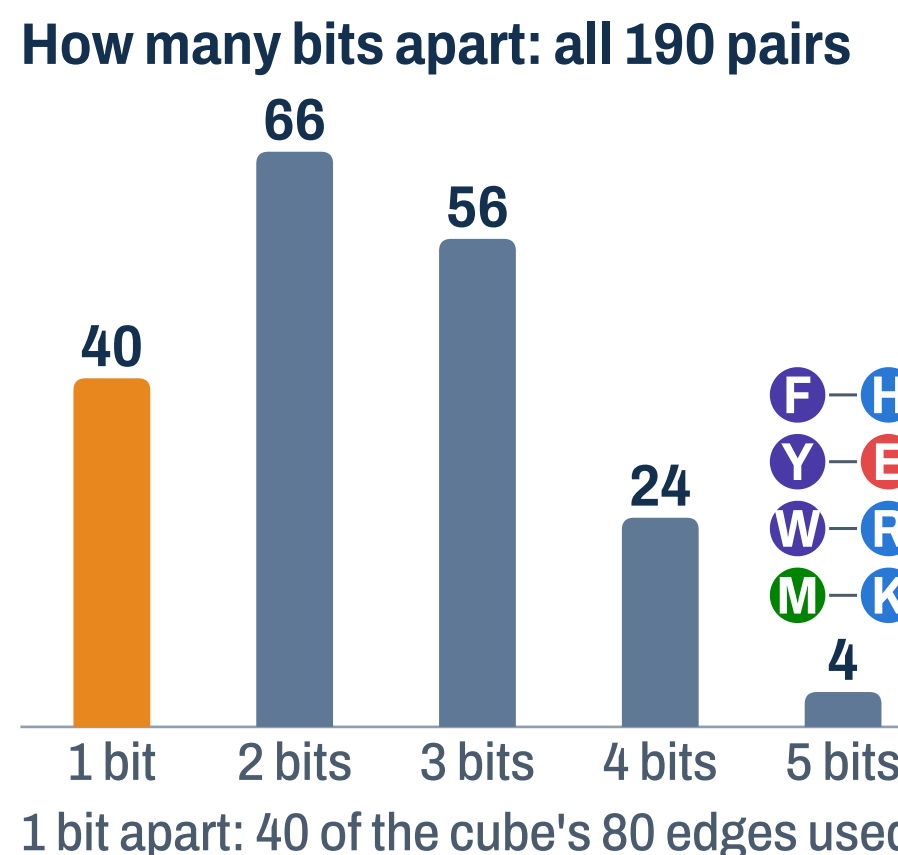
Flattened into a Karnaugh map: touching cells differ in one bit, and the edges wrap

first 2 bits

last 3 bits

	000	001	011	010	110	111	101	100	
00	A	V	L	I	F	Y	W	M	
01	D	Q	N	T	S	G	P	C	
11	E	H	K	R					
10									

- aliphatic hydrophobic
- aromatic
- sulfur-containing
- structure-breaking
- polar uncharged
- negatively charged
- positively charged
- unused codeword (12)



110 Contacts as a Boolean function

A contact map is a Boolean function: $C(i,j) = 1$ when residues i and j lie within 8 Å. Quine–McCluskey⁵ merges 1-cells one bit apart into ever larger cubes; the cubes that cannot grow are the prime implicants, and a cover of them is a two-level circuit equal to the map. For an 8-residue example Lean proves the map symmetric and that no two of its contacts can merge.

Merge cells one bit apart

8 residues, 4 contacts

AB 00 01 11 10 CD	
00	0101+0111 $\rightarrow$ 01-1
01	1101+1111 $\rightarrow$ 11-1
11	01-1+11-1 $\rightarrow$ -1-1
10	1000+1001 $\rightarrow$ 100-

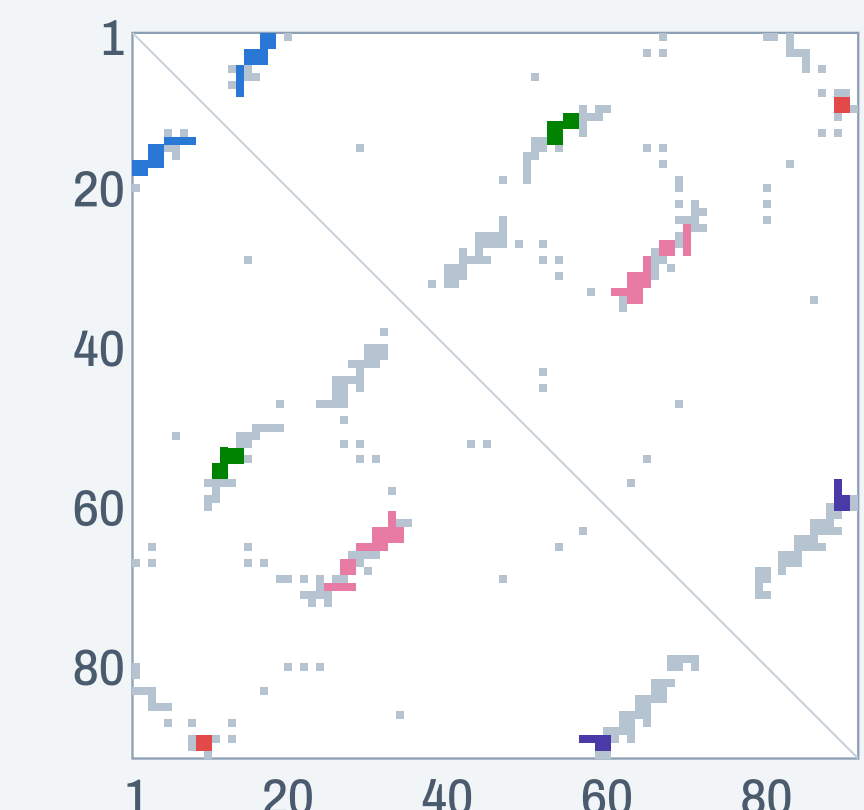
f = BD + ABC

dashed: prime, not needed

no two touch, so each is its own prime implicant

100 Rules with no exception, from exact circuits

Each PSICOV⁷ protein becomes a circuit equal to its contact map. Its AND gates are rules; block rules that continue one another give strand pairs, direction and register, by counting.



One AND gate, read as a rule

i 001101 - IF 27–28 AND 67–68

j 100001 - THEN contact (0 exceptions)

Read from the rules: strand pairs

- 25–34 $\cdot$ 61–70 antiparallel $i + j = 94-98$
- 1–8 $\cdot$ 14–18 antiparallel hairpin $i + j = 18-22$
- 11–14 $\cdot$ 53–56 antiparallel $i + j = 65-68$
- 57–60 $\cdot$ 89–90 parallel $j - i = 29-32$

150 / 150 proteins exact on all 6,905,856 inputs
27,997 rules, 0 exceptions, 0 contacts missed
378 antiparallel, 215 parallel pairs, 104 hairpins

One species: in 1,299 SARS-CoV-2 Spike sequences, exact circuits list which residue pairs occur at co-evolving positions and which never do. No sequence breaks a rule.

positions 371 $\times$ 376
1,294 sequences

A376	T376
F371	970 0
L371	0 311
S371	1 12

F371 $\Rightarrow$ A376 in 970 of 970
L371 $\Rightarrow$ T376 in 311 of 311

positions 405 $\times$ 408
1,289 sequences

R408	S408
D405	319 0
N405	12 958

D405 $\Rightarrow$ R408 in 319 of 319
S408 $\Rightarrow$ N405 in 958 of 958

positions 486 $\times$ 493
1,297 sequences

Q493	R493
F486	20 1,173
P486	2 0
V486	102 0

V486 $\Rightarrow$ Q493 in 102 of 102
R493 $\Rightarrow$ F486 in 1,173 of 1,173

10 position pairs $\cdot$ 41 allowed and 35 forbidden residue combinations $\cdot$ 23 implications
0 violations: every one of the 1,299 sequences obeys every rule

Takeaways

- A verified foundation. Lean fixes the encodings and what a block rule means; every circuit is exact on all inputs.
- Rules that are true, not likely. Each holds for every contact, sequence or frame it names.
- One language for sequence, structure and dynamics. Any finite encoding can be checked the same way.

Try any PDB entry, from file to circuit, rules and inferences:
e-motioner-x-sbs.github.io/ICMS_Submission

111 How Lean checks a claim

In Lean 4⁶ a statement is a type and a proof is a program checked by a small kernel. For finite claims, native_decide evaluates every case in compiled code, up to 2²⁰ in one theorem, and so also trusts Lean's compiler. No proof uses sorry or adds an axiom of its own.

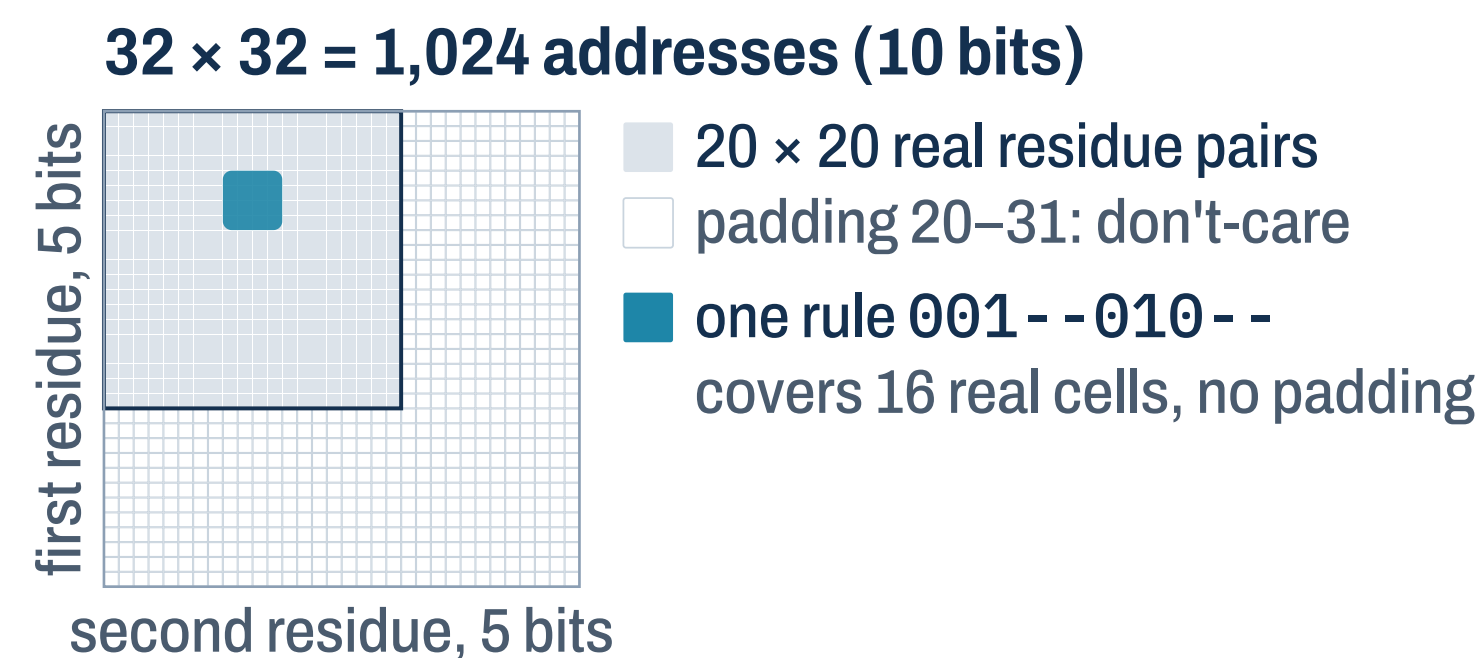
```
-- Gray code is additive over GF(2)
theorem sc_gray_gf2_linear :
  ∀ x y : Fin 32,
    scGray (x.val ^^^ y.val)
      = scGray x.val ^^^ scGray y.val
:= by native_decide
```

native_decide	standard axioms only	no axiom
KmapProofs	16	
KmerIndexing	35	
AminoAcidEncoding	27	
ContactMapCompleteness	20	
KmapEncodingEquiv	8	
ContactCircuits	12	131 theorems
SequenceCircuits	13	100 + 23 + 8

101 What the proofs guarantee

Exact rules on padded maps

Twenty residues need five bits per axis, so a 20 $\times$ 20 residue-pair map is padded to 32 $\times$ 32 with 10-bit addresses. Lean proves that a fully fixed rule matches exactly its own cell and that such a rule on two real residues fires only inside the 20 $\times$ 20 block, never on padding.



Segments hold under either numbering

With its low position bits free, a cube is an unbroken chain segment. Lean proves this for Gray-coded positions as well as plain binary: the two numberings name different segments, never broken ones.

Same cube, still a segment, different residues

Plain binary cube 10 - - $\rightarrow$ residues 8–11

00 00 00 00 01 01 01 01 10 10 10 10 11 11 11 11

00 01 10 11 00 01 10 11 00 01 10 11 00 01 10 11

0-1-2-3-4-5-6-7-8-9-10-11-12-13-14-15

00 00 00 00 01 01 01 01 11 11 11 11 10 10 10 10

00 01 11 10 10 11 01 00 00 01 11 10 10 11 01 00

Gray-coded cube 10 - - $\rightarrow$ residues 12–15

Lexicographic (FCGR) Gray Same 16 counts, moved

A C G T $i \rightarrow g(i)$ A C T G CGCGAATTCGCG

A 1 1 00 $\rightarrow$ 00 1 1 A (1BNA), address order

C 1 4 01 $\rightarrow$ 01 4 C

G 1 2 10 $\rightarrow$ 10 1 T

T 1 1 11 $\rightarrow$ 11 12 G

◆ cophenetic r
0.976747 = 0.976747

What Gray changes: code distance d , 400 residue pairs

20 80 132 112 48 8

$d=0$ $d=1$ $d=2$ $d=3$ $d=4$ $d=5$

80 / 400 pairs one bit apart

Crambin, 8,000 frames
a = contact 4–44, t = 4–43

t formed	t broken
a formed 5,429	0
a broken 1,258	1,313

9,986 rules hold in every frame of 479 folding runs
like $a \Rightarrow t$: a formed never without t

3,966 join neighbouring contacts 33 / 33 in the long runs

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2. H. J. Jeffrey, *Nucleic Acids Res.* **1990**, 18, 2163–2170.
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